Parenthood Timing and Gender Inequality

Julius Ilciukas

The Chinese University of Hong Kong

Motivation

Gender inequality in OECD labor markets typically emerges with parenthood

- Understanding the role of parenthood is crucial for addressing gender inequity

Quantifying the effects of parenthood is challenging due to selection:

- Parents may differ systematically from those who are and/or remain childless

Addressing selection requires quasi-experimental fertility variation:

- A setting where pregnancy is as-good-as-random (various quasi-experiments)
- Compare those who conceive at that moment to those who do not

But many women who do not conceive initially eventually become mothers

- Existing evidence identifies a weighted average of parenthood and timing effects

These forces may work in opposite directions and require distinct policies

This Paper

What are the effects of parenthood and its timing?

- How would outcomes differ if individuals remained childless or delayed childbirth?

1. New approach to estimate treatment effects in sequential quasi-experiments
   - Settings where previously unassigned individuals can repeatedly undergo assignment
   - E.g. a series of conception attempts
2. Empirical evidence using novel Dutch administrative data
   - Focus on couples undergoing intrauterine insemination
   - Common and minimally invasive infertility treatment
3. Quantify and compare the effects of parenthood and its timing
   - How does parenthood shape women's outcomes and gender inequality?
   - Does later childbearing mitigate or amplify losses?

Literature and Contribution

1. Existing quasi-experimental studies average parenthood and timing effects
   - Hotz et al. (2005); Agüero & Marks (2008); Cristia (2008); Miller (2011); Lundborg et al. (2017); Bensnes et al. (2023); Gallen et al. (2023); Lundborg et al. (2024)

I provide the first estimates that separately quantify these effects

Literature and Contribution

1. Existing quasi-experimental studies average parenthood and timing effects
   - Hotz et al. (2005); Agüero & Marks (2008); Cristia (2008); Miller (2011); Lundborg et al. (2017); Bensnes et al. (2023); Gallen et al. (2023); Lundborg et al. (2024)

I provide the first estimates that separately quantify these effects

2. The most credible evidence comes from unique samples and uses Scandinavian data
   - Lundborg et al. (2017); Bensnes et al. (2023); Gallen et al. (2023); Lundborg et al. (2024)

I study a common, minimally invasive infertility treatment using Dutch data

Literature and Contribution

1. Existing quasi-experimental studies average parenthood and timing effects
   - Hotz et al. (2005); Agüero & Marks (2008); Cristia (2008); Miller (2011); Lundborg et al. (2017); Bensnes et al. (2023); Gallen et al. (2023); Lundborg et al. (2024)

I provide the first estimates that separately quantify these effects

2. The most credible evidence comes from unique samples and uses Scandinavian data
   - Lundborg et al. (2017); Bensnes et al. (2023); Gallen et al. (2023); Lundborg et al. (2024)

I study a common, minimally invasive infertility treatment using Dutch data

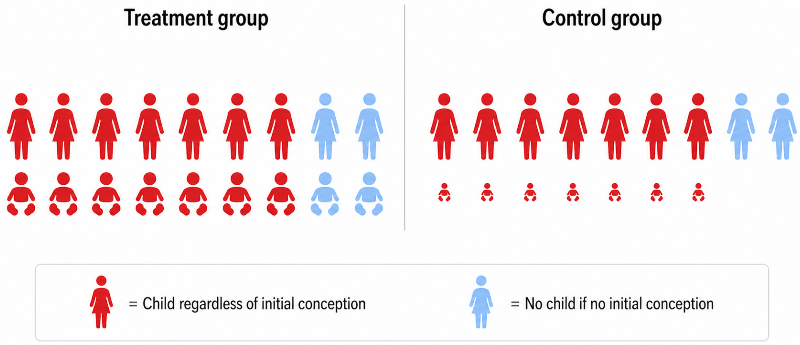
3. Separating treatment and timing effects in quasi-experiments is a common challenge
   - Education programs with multiple admission cycles, assignment to judges, promotion tournaments

Method applicable to many other settings with sequential quasi-experiments

Typical Quasi-Experiment

At a given moment, some women conceive (treatment), others do not (control)

- Control group: remain childless or conceive later



Mothers differ in career stage at birth, motherhood duration, and potentially # children

- Comparison mixes motherhood vs. childlessness with earlier vs. later motherhood

Baseline Reduced-Form Setup

Address selection using a quasi-experiment: women undergoing artificial insemination.

- Outcome of the first procedure $Z_1 \in \{0, 1\}$
- Parenthood indicator $D \in \{0, 1\}$
- Potential labor market outcomes $Y_{z_1}(d)$
- Effect of parenthood: $Y_1(1) - Y_0(0)$
- Effect of parenthood timing: $Y_1(1) - Y_0(1)$

Baseline Reduced-Form Setup

Address selection using a quasi-experiment: women undergoing artificial insemination.

- Outcome of the first procedure $Z_1 \in \{0, 1\}$
- Parenthood indicator $D \in \{0, 1\}$
- Potential labor market outcomes $Y_{z_1}(d)$
- Effect of parenthood: $Y_1(1) - Y_0(0)$
- Effect of parenthood timing: $Y_1(1) - Y_0(1)$

Standard reduced form argument:

1. Assume success of the first procedure is random (conditional on age)
2. Compare women whose first procedure succeeds to those whose fails:

$$\mathbb{E}[\text{Parenthood effect} \mid D = 0 \text{ if } Z_1 = 0]c + \mathbb{E}[\text{Timing effect} \mid D = 1 \text{ if } Z_1 = 0](1 - c)$$

- $c = \Pr(\text{no child if } Z_1 = 0) \approx 0.25$ in practice

Baseline Reduced-Form Setup

Address selection using a quasi-experiment: women undergoing artificial insemination.

- Outcome of the first procedure $Z_1 \in \{0,1\}$
- Parenthood indicator $D \in \{0,1\}$
- Potential labor market outcomes $Y_{z_1}(d)$
- Effect of parenthood: $Y_1(1) - Y_0(0)$
- Effect of parenthood timing: $Y_1(1) - Y_0(1)$

Standard reduced form argument:

1. Assume success of the first procedure is random (conditional on age)
2. Compare women whose first procedure succeeds to those whose fails:

$$\mathbb{E}[\text{Parenthood effect} \mid D = 0 \text{ if } Z_1 = 0]c + \mathbb{E}[\text{Timing effect} \mid D = 1 \text{ if } Z_1 = 0](1 - c)$$

- $c = \Pr(\text{no child if } Z_1 = 0) \approx 0.25$ in practice

Consequently: if Z_1 used as IV for $D \rightarrow$ biased estimates of parenthood effect IV methods

Method Overview

Key observation: many women conceive through subsequent IUIs.

- Variation in later IUI success could help separate treatment and timing effects
- Leveraging this variation is challenging because the number of IUIs is endogenous

Result 1: Selection into subsequent IUIs can be addressed using the observed IUI #

- Identify average outcomes in the scenario where entire IUI sequence fails
- Many women remain childless; some conceive (much later) via natural means

Result 2: Worst-case bounds allow to separate treatment and timing effects

- Bound the effect of conceiving at the first IUI vs. remaining childless
- Bound the effect of conceiving at the first IUI vs. after the IUI sequence

No assumptions required on selection into subsequent IUIs and natural conception or on the structure of treatment/timing effects within or across individuals.

Statistical Model

Women differ in two unobserved characteristics:

- “Willingness” to undergo IUIs, $W \in \{1, \ldots, \overline{w}\}$
  - Would undergo W IUIs for the first child if all IUIs failed
- “Reliance” on IUIs, $R \in \{0, 1\}$
  - No child if all IUIs fail, $R = 1$

Statistical Model

Women differ in two unobserved characteristics:

- "Willingness" to undergo IUIs, $W \in \{1, \ldots, \overline{w}\}$
  - Would undergo W IUIs for the first child if all IUIs failed
- "Reliance" on IUIs, $R \in \{0, 1\}$
  - No child if all IUIs fail, $R = 1$

Observables:

- IUI j success indicator, Z_j, for procedures before having any children
- Number of realized IUIs:

$$A = \min\left(\{j : Z_j = 1\} \cup \{W\}\right)$$

- Parenthood indicator:

$$D = \max(Z_A, 1 - R)$$

Sequential Unconfoundedness

Assumption (Sequential Unconfoundedness)

$(Y_1(1), Y_0(0), R, W) \perp\!\!\!\perp Z_j | A \geq j$.

In words: once sperm at IUI j are implanted, whether this results in a conception is as-good-as-random

- **The decision to undergo the procedure can be endogenous**
- $Y_1(1), Y_0(0), R$ and W can be related
- Main method relaxes to covariate-conditional version: age at procedure, technology

Simple World: Max 2 IUIs, All Reliers

$W = 1$

(willing to try once)

$Z_1 = 1$

$Z_1 = 0$

Simple World: Max 2 IUIs, All Reliers

$W = 1$
(willing to try once)

$Z_1 = 1$

$Z_1 = 0$

$W = 2$
(willing to try twice)

$Z_1 = 1$

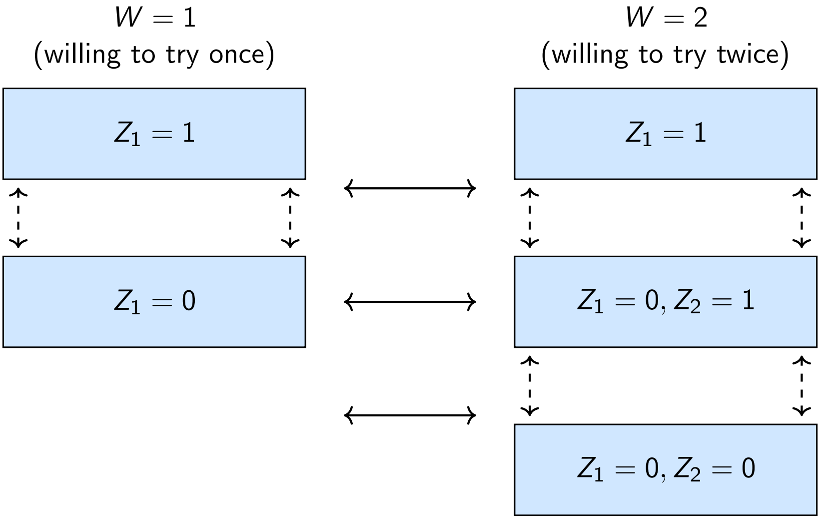
$Z_1 = 0, Z_2 = 1$

$Z_1 = 0, Z_2 = 0$

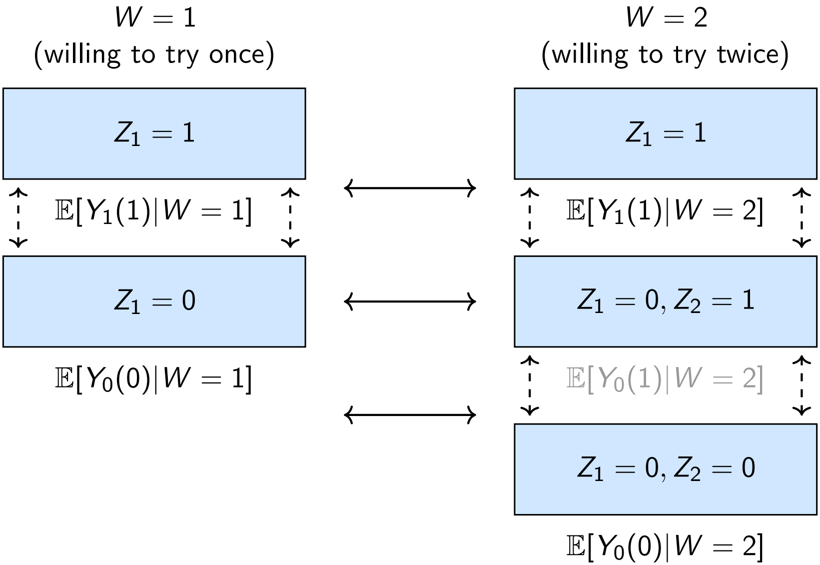
Simple World: Max 2 IUIs, All Reliers

$W = 1$ (willing to try once)		$W = 2$ (willing to try twice)
$Z_1 = 1$	$\longleftrightarrow$	$Z_1 = 1$
$Z_1 = 0$	$\longleftrightarrow$	$Z_1 = 0, Z_2 = 1$
	$\longleftrightarrow$	$Z_1 = 0, Z_2 = 0$

Simple World: Max 2 IUIs, All Reliers



Simple World: Max 2 IUIs, All Reliers



Simple World (Observed): Max 2 IUIs, All Reliers

$W = 1$
(willing to try once)

$W = 2$
(willing to try twice)

$Z_1 = 1$

$Z_1 = 0$

$Z_1 = 0, Z_2 = 1$

$Z_1 = 0, Z_2 = 0$

Simple World (Observed): Max 2 IUIs, All Reliers

$W = 1$ (willing to try once)	$W = 2$ (willing to try twice)

$Z_1 = 1$

$\mathbb{E}[Y_1(1)]$

$Z_1 = 0$

$\mathbb{E}[Y_0(0)|W = 1]$

$Z_1 = 0, Z_2 = 1$

$\mathbb{E}[Y_0(1)|W = 2]$

$Z_1 = 0, Z_2 = 0$

$\mathbb{E}[Y_0(0)|W = 2]$

Simple World (Observed): Max 2 IUIs, All Reliers

$W = 1$
(willing to try once)

$W = 2$
(willing to try twice)

$Z_1 = 1$

$\mathbb{E}[Y_1(1)]$

$Z_1 = 0$

$\mathbb{E}[Y_0(0)|W = 1]$

$Z_1 = 0, Z_2 = 1$

$\mathbb{E}[Y_0(1)|W = 2]$

$Z_1 = 0, Z_2 = 0$

$\mathbb{E}[Y_0(0)|W = 2]$

Simple World (Observed): Max 2 IUIs, All Reliers

$W = 1$
(willing to try once)

$W = 2$
(willing to try twice)

$Z_1 = 1$

$\mathbb{E}[Y_1(1)]$

$Z_1 = 0$

$\mathbb{E}[Y_0(0)|W = 1]$

$Z_1 = 0, Z_2 = 1$

$\mathbb{E}[Y_0(1)|W = 2]$

$Z_1 = 0, Z_2 = 0$

$\mathbb{E}[Y_0(0)|W = 2]$

$Pr(W = 1) =$

Simple World: Max 1 IUI with Non-reliers

$R = 1$ (no child if fail)	$R = 0$ (child if fail)

Simple World: Max 1 IUI with Non-reliers

$R = 1$ (no child if fail)	$R = 0$ (child if fail)
$Z_1 = 1$	

Simple World: Max 1 IUI with Non-reliers

$R = 1$ (no child if fail)	$R = 0$ (child if fail)
$Z_1 = 1$	$Z_1 = 1$
$Z_1 = 0, D = 0$	$Z_1 = 0, D = 1$

Simple World: Max 1 IUI with Non-reliers

$R = 1$
(no child if fail)

$R = 0$
(child if fail)

$Z_1 = 1$

Distribution of $Y_1(1)$

$Z_1 = 0, D = 0$

$Z_1 = 0, D = 1$

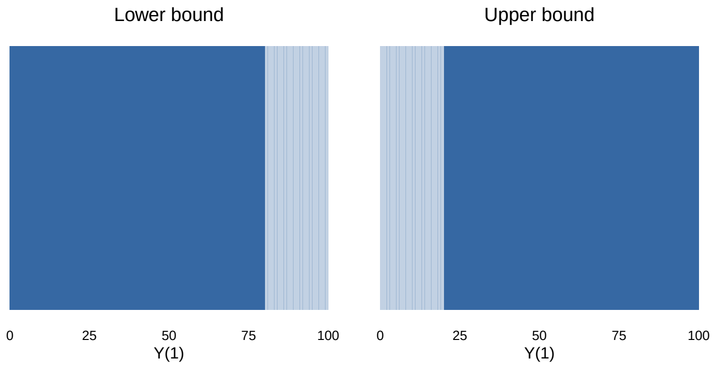
$\mathbb{E}[Y_0(0)|R = 1]$

$\mathbb{E}[Y_0(1)|R = 0]$

$Pr(R = 1) =$

Intuition: Motherhood Outcome $Y_1(1)$

1. Treated group is a representative sample but their types are unobserved
2. Identify $\Pr(R = 1) = 0.8$ on control group
3. Assume most extreme distributions of types in treated group
4. Bound $\mathbb{E}[Y_1(1)|R = 1]$



Argument for timing effects is symmetric.

Background and Data

Assisted conception procedures →

- In-vitro fertilization: invasive medical procedure, first 3 free
- Intrauterine insemination: direct sperm injection, minimally invasive, free

Dutch family policies and labor market similar to OECD average →

- 16 weeks maternity + pregnancy leave, 1 week paternity leave

Data combining ACP medical records with tax records →

- Work hours and income include leave; results for hours corrected for uncertainty
- 15,523 cohabiting opposite-sex couples
- Balance: ACP success at each attempt uncorr. with past outcomes cond. on age
- Reliers make up $\sim 50\%$ of the sample

Balance in 1st IUI first | Balance in later IUIs | Success probabilities per attempt

Parenthood and Timing Effect Estimates

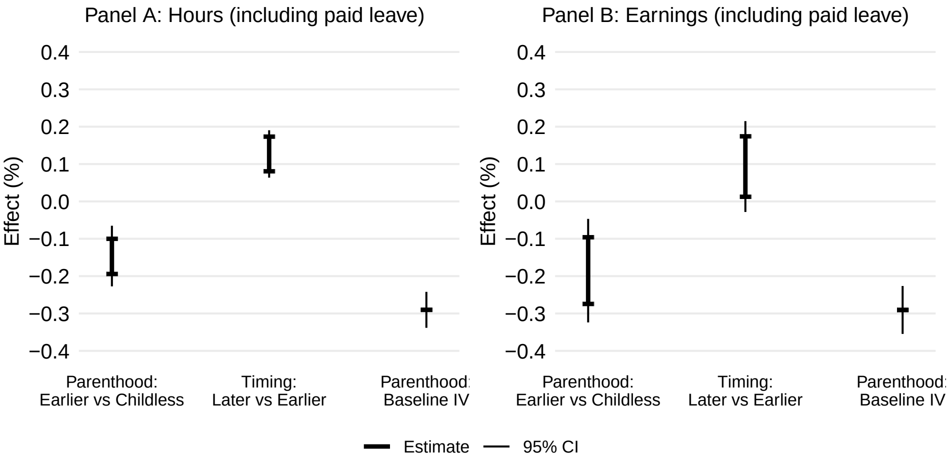
Main outcomes:

- Women's total earnings and work hours over the first seven years from first IUI

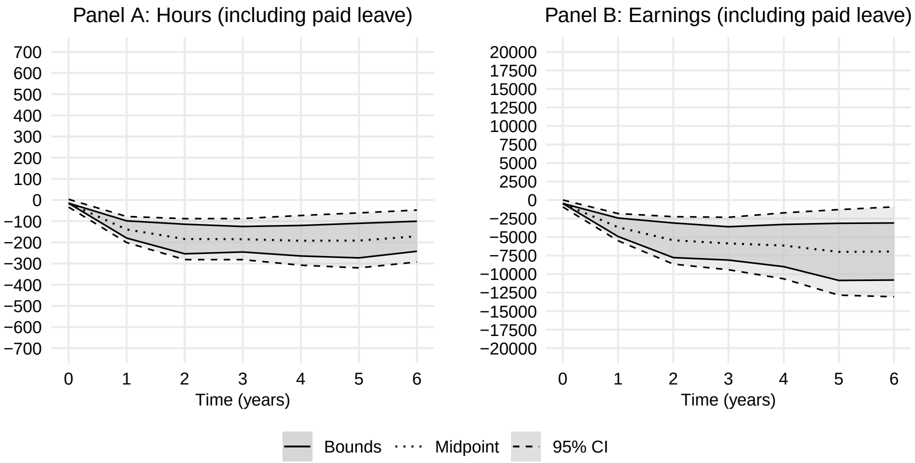
Main estimates:

1. Parenthood effect: effect of conception at first IUI relative to remaining childless
2. Timing effect: birth after IUI sequence failure relative to conception at first IUI
   - Effect of average delay of ~ 3 years
   - Combines effects before and after the delayed birth
3. Naive IV parenthood effect estimates that assume no timing effects

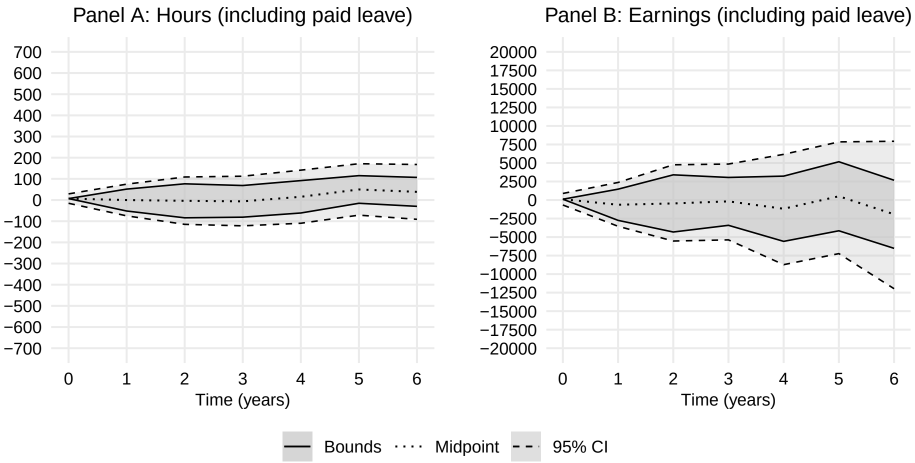
Effect of Motherhood



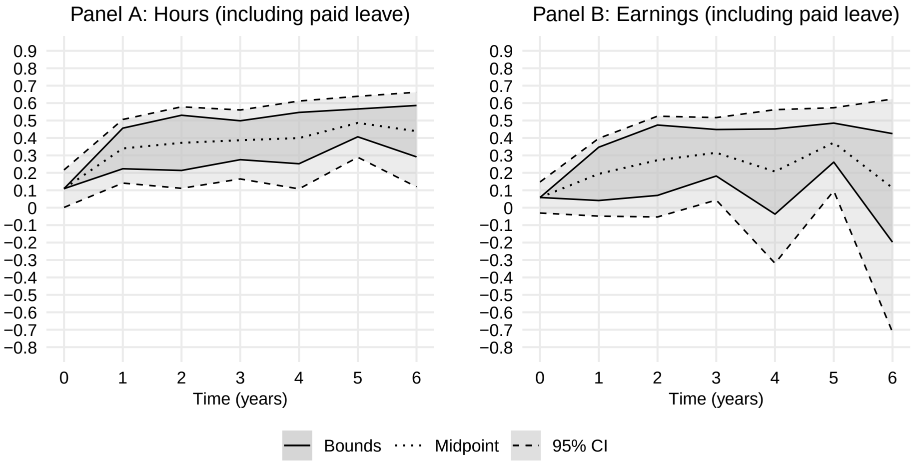
Dynamic Effect of Motherhood



Dynamic Effect of Fatherhood



Share of Gender Inequality Caused by Parenthood



Effect on gender gap relative to gap Under parenthood

Decomposing Timing Effects

On average, women whose IUIs fail conceive 3.1 years later.

Estimates of total effects over 7 years from first IUI combine two periods:

1. Only the treatment group has children
2. Both groups have children, but from different moments

To separate the two I examine cumulative effects over the first t years from first IUI.

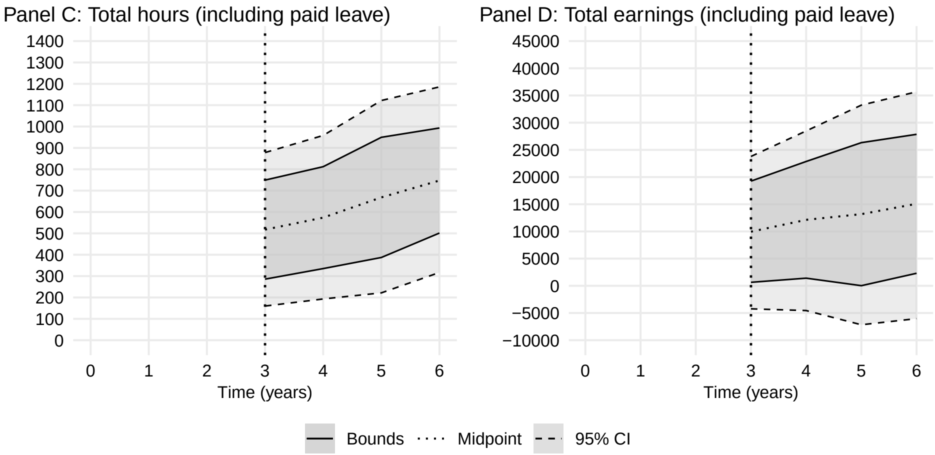
Since most delayed births occur $\sim$ 3 years later:

(a) At year 3: cumulative effect of no child vs. a child conceived 3 years ago

(b) At year $3 + t$: cumulative effect of motherhood starting $3 + t$ vs. t years ago

Difference (b) – (a): after both groups become mothers, do women who enter motherhood later perform better or worse than those who enter earlier?

Cumulative Effect of Motherhood Timing



How Much Does Delayed Childbearing Mitigate Career Costs?

Focus on midpoints of bounds

- Necessarily suggestive; comparing short run estimates for reliers and non-reliers

Before delayed birth:

- Absence of children: +16% annual hours, +19% earnings

After delayed birth:

- Gains shrink to +5% hours, +2% earnings

Lifetime implications (33-year horizon from age 32)

- 1-year delay $\Rightarrow$ +5.5% hours, +2.5% earnings
- 5-year delay $\Rightarrow$ +7% hours, +4.5% earnings

Baseline long-run parenthood losses: –15% hours, –20% earnings

Five-year delay reduces losses to: –9% hours, –16% earnings

Conclusion

Results:

- Motherhood accounts for up to half of gender inequality in hours and income
- Delayed childbearing mitigates women's losses

External relevance:

- IUI is common; the sample matches the population on observables
- Many women would undergo IUI if natural conception failed
- Parenthood effects exceed Scandinavian IVF/contraceptive-failure estimates

Policy:

- Enabling later childbearing may mitigate gender inequality
- Some gaps might persist even in the absence of children

Extensions

- **Selection into parenthood** Formal procedure Estimates
- Relation to partial identification literature Comparison
- Robustness Childless final period De-aging partners Mental health Bounds for non-depressed

How Existing Work Isolates Parenthood Effects

Parenthood effects may depend on current lifecycle stage and duration of motherhood:

$$\tau(\text{age}, \text{duration})$$

- Age $=$ current career stage
- Duration $=$ first child age
- **Age x Duration** $=$ career stage at first birth

A) Standard IV approach (Lundborg et al., 2017, 2024):

$$\tau(\text{age}, \text{duration}) = \tau(\text{age})$$

B) Recursive IV approach (Bensnes et al., 2023; Gallen et al., 2023):

$$\tau(\text{age}, \text{duration}) = \tau(\text{duration}) \quad \text{(and homogeneous effects)}$$

- A) requires that duration does not matter, B) that career stage does not matter
- Both A) and B) require that **career stage at parenthood** does not matter
- Estimates differ substantially across approaches (Bensnes et al., 2023)

Back

ToC

Bounding τ_{ATR}

Construct the moment:

$$m^L(G,\eta^0) = Y 1_{\{Y<q(r(X_1),X_1)\}} \frac{Z_1}{e_1(X_1)} - Y(1-D)\prod_{j=1}^{A}\frac{(1-Z_j)}{(1-e_j(X_j))}$$

- G is the observed data vector
- η^0 contains the following:
  - $e_j(X_j) = \Pr(Z_j = 1|X_j)$
  - $q(r(X_1), X_1)$ is the $r(X_1)$-th quantile of Y given X_1 and $Z_1 = 1$
  - $r(X_1)$ identifies the covariate-conditional relier share

Assumption (Conditional Sequential Unconfoundedness)

$(Y(k), R, W) \perp\!\!\!\perp Z_j \mid X_j$ *for all* j, k, *and* $X_j, A \geq j$.

Theorem (Lower Bound)

Under conditional sequential unconfoundedness and regularity, the sharp lower bound on τ_{ATR} *is* $\mathbb{E}[m^L(G,\eta^0)]/\mathbb{E}[r(X_1)]$.

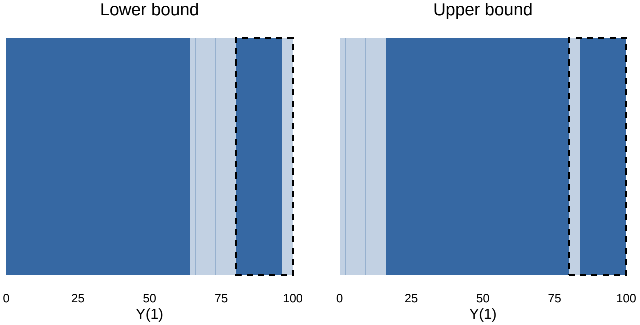
Back

ToC

Intuition: Motherhood Outcome $Y_1(1)$—Covariates

Pre-IUI covariates can help narrow the bounds:

- Can identify relier share at each covariate value
- Baseline bounds assume extreme scenarios where reliers have highest or lowest treated outcomes
- These distributions of treated outcomes might be inconsistent with conditional relier shares



Estimating the Bounds

Distribution of $m^L(G, \eta^0)$ is complicated by $q(r(X_1), X_1)$

- Semenova (2023) addresses a closely related inference challenge
- Double/debiased machine learning approach
  1. Adjust $m^L(G, \eta^0)$ to make it insensitive to small error in $q(r(X_1), X_1)$
  2. Sample splitting
- Asymptotic inference as if $q(r(X_1), X_1)$ was known

New moment:

$$\psi^L(G, \xi^0) = m^L(G, \eta^0) + corr(G, \xi^0)$$

Identifies same parameter:

$$\mathbb{E}[\psi^L(G, \xi^0)] = \mathbb{E}[m^L(G, \eta^0)]$$

Insensitive to estimation error in $q(r(X_1), X_1)$:

$$\partial_{q(.)} \mathbb{E}[\psi^{L+}(G, \xi_r)|X_1]|_{\xi_r = \xi_r^0} = 0 \ a.s.$$

ToC Back

Assisted Conception Procedures

- IUI (main procedure): sperm injected into uterus
  - Minimally invasive, primary ACP in most countries
  - “Free” in NL
- IVF (secondary procedure): embryo inserted into uterus
  - Invasive treatment, performed under sedation/anesthesia
  - Eggs retrieved through the vaginal wall using a specialized needle
  - In NL, first 3 free; each subsequent costs between 1000 and 4000 EUR

Back

ToC

Institutions

- Dutch family friendly polices similar to OECD average
  - 16 weeks of fully paid pregnancy+maternity leave
  - 1 week of paternity leave
  - Average time in child care similar to OECD average
  - Net child care cost 10% median household income
- Dutch employment intensity similar to OECD average
  - Employment among parents and non-parents relatively high
  - Part time work much more common
  - Approximately 15% two-parent families have both partners working part-time

Back

Data

Administrative data from Statistics Netherlands

- Comprehensive hospital records cover fertility treatments from 2012 to 2017: procedure date and type
  - Success imputed as having child born within 10 months
- Tax records cover work hours and income from 2011 to 2023
  - Include maternity leave and pay
  - Main bounds account for uncertainty around actual work hours
- Birth dates, legal family connections, cohabitation
- Dispensed medication registry

Main sample: cohabiting opposite-sex couples undergoing IUI for their first child between 2013 and 2016: 15,523

Back

Balance in 1st IUI

Back (data) Back (representativeness)

Table 1: First IUI Outcomes and Descriptives

	Success (1)	Fail (2)	Dif. (1)-(2)	IPW dif. (1)-(2) cond.	Rep. (5)	Suc. vs rep. (1)-(5)
Work (W)	0.912	0.916	-0.004	-0.009	0.936	-0.024
	[0.283]	[0.277]	(0.008)	(0.008)	[0.244]	(0.007)
Work (P)	0.894	0.885	0.009	0.002	0.897	-0.002
	[0.307]	[0.319]	(0.009)	(0.009)	[0.304]	(0.008)
Hours (W)	1300.012	1298.876	1.136	-1.951	1310.923	-10.911
	[547.832]	[558.316]	(15.730)	(16.119)	[544.468]	(14.554)
Hours (P)	1513.337	1494.541	18.796	3.345	1497.603	15.734
	[635.121]	[656.050]	(18.457)	(19.041)	[651.043]	(17.403)
Earn. 1000s EUR (W)	29.358	29.648	-0.290	0.203	26.555	2.803
	[18.000]	[18.911]	(0.531)	(0.561)	[15.989]	(0.427)
Earn. 1000s EUR (P)	38.082	38.060	0.022	0.322	33.862	4.220
	[25.425]	[26.525]	(0.745)	(0.774)	[24.148]	(0.646)
Bachelor deg. (W)	0.512	0.494	0.018		0.518	-0.007
	[0.500]	[0.500]	(0.014)		[0.500]	(0.013)
Bachelor deg. (P)	0.425	0.410	0.014		0.430	-0.005
	[0.494]	[0.492]	(0.014)		[0.495]	(0.013)
Age (W)	31.373	32.060	-0.687		28.840	2.533
	[3.889]	[4.265]	(0.119)		[3.896]	(0.104)
Age (P)	34.088	34.856	-0.768		31.415	2.673
	[4.968]	[5.500]	(0.154)		[4.803]	(0.128)
Observations	1,411	11,323			171,180	
Joint p-val.			0.001	0.955		0.000

Note: *Success* – average among women whose first IUI succeeded; *Fail* – average among women whose first IUI failed; *Dif.* – difference between *Success* and *Fail*; *IPW dif.* – difference adjusted for age and education using inverse probability weights from the baseline specification; *Rep.* – average in representative sample of women who conceived their first child without assisted conception procedures; *Suc. vs rep* – difference between *Success* and *Rep.*. Reference year: year of first IUI (IUI sample); 9 months before first birth (representative sample). IUI sample: women who underwent intrauterine insemination for their first child between 2013 and 2016, with no prior assisted conception procedures, cohabiting with a male partner in the year prior to the reference year. Representative sample: women with no assisted conception procedures before first birth, cohabiting with a male partner in the year prior to the reference year, with reference year between 2013 and 2016. Labor market outcomes measured in the year before the reference year; age measured in the reference year. *Bachelor deg.* – indicator for completing a bachelor's degree. *Earn.* – earnings, (W) – woman, (P) – partner. Standard deviations in brackets. Standard errors in parentheses.

ToC

Balance in Subsequent ACPs

Table 2: Balance in Later ACPs

	Z_2	Z_3	Z_4	Z_5	Z_6	Z_7	Z_8	Z_9	Z_{10}
Work (W)	0.009	-0.004	0.022	0.014	0.039	-0.003	-0.011	0.022	0.030
	(0.010)	(0.011)	(0.011)	(0.012)	(0.012)	(0.017)	(0.018)	(0.019)	(0.024)
Work (P)	0.006	0.016	0.012	0.020	-0.004	-0.004	-0.019	0.017	0.030
	(0.010)	(0.010)	(0.012)	(0.012)	(0.015)	(0.015)	(0.019)	(0.020)	(0.027)
Hours (W)	32.885	-4.482	52.999	41.332	81.957	11.894	-18.836	72.659	24.819
	(18.721)	(20.032)	(21.045)	(22.686)	(25.131)	(31.187)	(32.937)	(38.210)	(48.490)
Hours (P)	21.655	24.730	23.756	38.965	9.666	-6.580	-28.458	30.525	43.722
	(21.018)	(21.089)	(23.574)	(25.255)	(30.585)	(31.513)	(37.976)	(44.856)	(52.821)
Income 1000s € (W)	1.481	-0.015	1.685	1.802	2.086	0.150	-0.043	0.866	-0.444
	(0.615)	(0.624)	(0.767)	(0.830)	(0.913)	(1.000)	(1.092)	(1.234)	(1.629)
Income 1000s € (P)	-0.749	1.002	2.040	0.800	0.774	0.025	0.259	-0.324	0.149
	(0.835)	(0.912)	(1.066)	(1.115)	(1.424)	(1.424)	(1.563)	(1.737)	(2.203)
Observations	12,974	10,774	8,726	6,977	5,411	3,944	2,723	1,850	1,174
Joint *p*-val.	0.175	0.976	0.234	0.303	0.140	1.000	0.956	0.704	0.917

Note: Each column describes the difference in average characteristics between women for whom the respective ACP succeeds and those for whom it fails, among those who undergo the procedure, using inverse probability weights for each ACP following the main specification. Labor market outcomes and age measured year before first treatment. (W) - woman, (P) - partner. Standard errors in parentheses.

Back

ToC

Estimated Success Probabilities

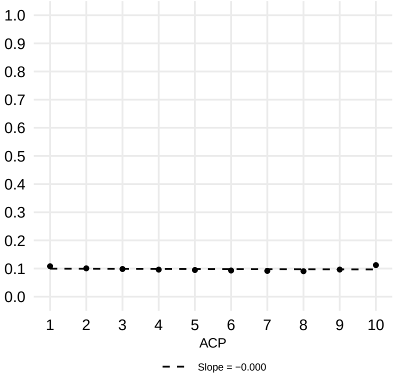


Figure 1: Estimated Success Probabilities

Back

ToC

Comparison with Lee (2009)

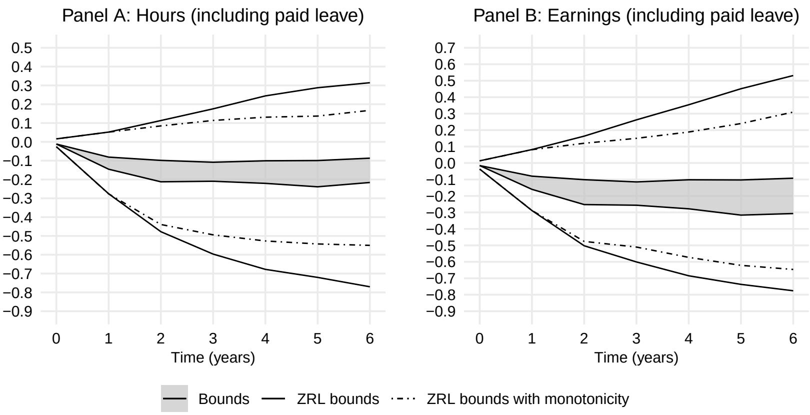


Figure 2: Comparison with Lee (2009) Bounds for Effects on Women

Quantifying Selection Using the Event Study Approach

Kleven et al. (2024) compare mothers t years after childbirth to women one year before

- Most gender inequality is explained by these differences between women
- Event study estimates may be biased due to selective timing
- But differences from IV/bounds need not imply bias, even in the same sample

1. I proxy fertility timing using the timing of first IUI
2. Compare the actual trajectories of women who attempted conception but remained childless with those imputed from women who are about to attempt conception but ultimately remain childless

- **Event study comparison of women with different timing absent children**
- **Same population as bounds: selection vs causal effects directly comparable**

Back (extensions) Conclusion

ToC

Placebo Event Study

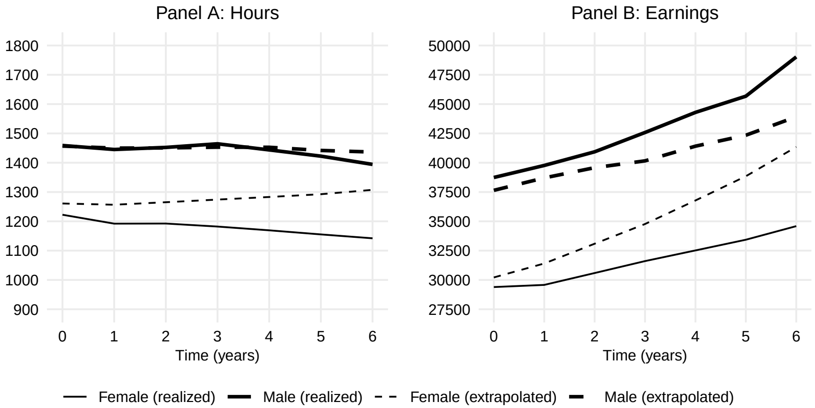


Figure 3: Placebo Event Study

Back (extensions) Conclusion

ToC

Gender Inequality: Causal vs Selection

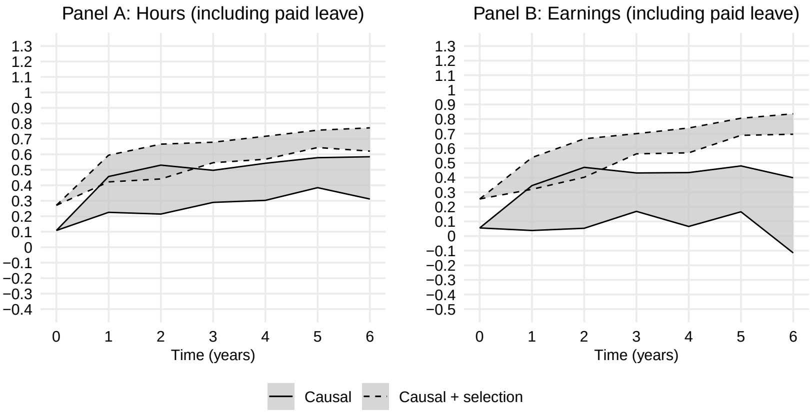


Figure 4: Share of Gender Inequality Explained by Selection and Parenthood

Back (extensions) Conclusion

ToC

Mental Health and Assisted Conception

Mental health consequences associated with failure to conceive are a part of the story:

- Unmet fertility goals may negatively impact mental health, and in turn, labor market outcomes

There are, however, additional concerns:

- Mental health issues caused specifically by failed conception or ACPs (external)
    - Focusing on artificial insemination helps mitigate this
- Large impacts unique to ACP families (external)
- Worsened mental health by threatening monotonicity (internal)

In practice, these impacts are likely small (Lundborg et al., 2024)

Antidepressant uptake | Back (extensions) | Conclusion

Monotone Bounds for Non-depressed Childless Women

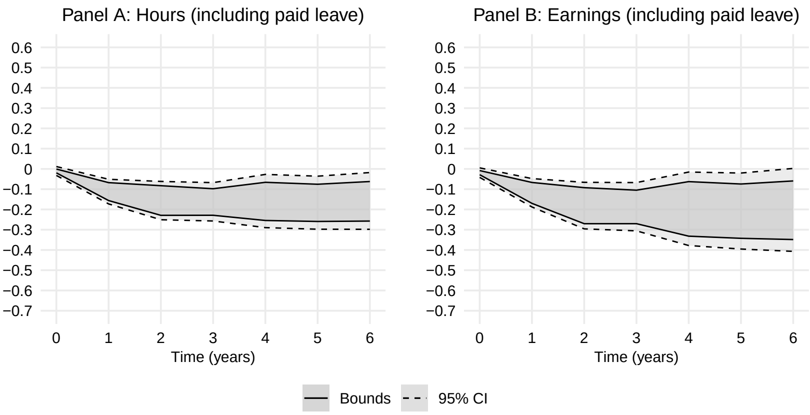


Figure 5: Monotone Bounds for Women Who Would Not Uptake Antidepressants if They Were to Remain Childless

Back (extensions) Conclusion

ToC

Effect of IUI Failure on Antidepressant Use

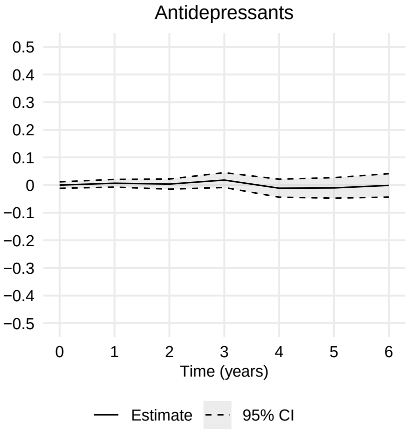


Figure 6: Estimates for effect on antidepressant take-up

Monotone Bounds: Women Who Remain Childless

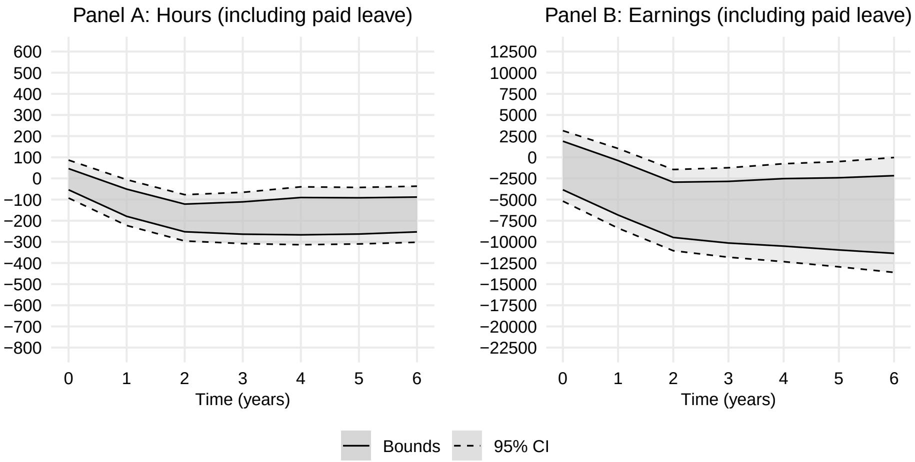


Figure 7: Monotone Bounds Using Completed Fertility

Parenthood Effect Bounds: Correcting for Partner's age

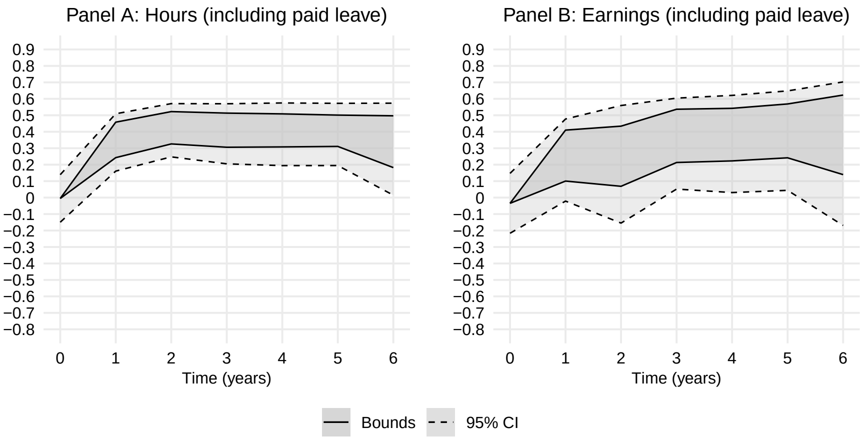


Figure 8: Bounds Measuring Male Income at Same Age as Female

References I

Agüero, J. M., & Marks, M. S. (2008). Motherhood and female labor force participation: evidence from infertility shocks. *American Economic Review*, *98*(2), 500–504.

Bensnes, S., Huitfeldt, I., & Leuven, E. (2023). Reconciling estimates of the long-term earnings effect of fertility. *IZA Discussion Paper*.

Cristia, J. P. (2008). The effect of a first child on female labor supply: Evidence from women seeking fertility services. *Journal of Human Resources*, *43*(3), 487–510.

Gallen, Y., Joensen, J. S., Johansen, E. R., & Veramendi, G. F. (2023). The labor market returns to delaying pregnancy. *Available at SSRN 4554407*.

Hotz, V. J., McElroy, S. W., & Sanders, S. G. (2005). Teenage childbearing and its life cycle consequences: Exploiting a natural experiment. *Journal of Human Resources*, *40*(3), 683–715.

Kleven, H., Landais, C., & Leite-Mariante, G. (2024). The child penalty atlas. *Review of Economic Studies*, rdae104.

Lee, D. S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *Review of Economic Studies*, *76*(3), 1071–1102.

Lundborg, P., Plug, E., & Rasmussen, A. W. (2017). Can women have children and a career? IV evidence from IVF treatments. *American Economic Review*, *107*(6), 1611–37.

Lundborg, P., Plug, E., & Rasmussen, A. W. (2024). Is there really a child penalty in the long run? new evidence from ivf treatments. *IZA Discussion Paper*.

Miller, A. R. (2011). The effects of motherhood timing on career path. *Journal of population economics*, *24*, 1071–1100.

Semenova, V. (2023). Generalized lee bounds. *arXiv preprint arXiv:2008.12720v3*.